string theory, these are string loops; in loop quantum gravity, they're harder to describe nonmathematically, but, roughly speaking, they're elementary loops of space—suggests there might be such a connection. This possibility is further supported by the fact that on the few problems accessible to both, such as black hole entropy, the two theories agree fully.[10] And, on the question of spacetime's constituents, both theories suggest that there is some kind of atomized structure. We've already seen the clues pointing toward this conclusion that arise from string theory; those coming from loop quantum gravity are compelling and even more explicit. Loop researchers have shown that numerous loops in loop quantum gravity can be interwoven, somewhat like tiny wool loops crocheted into a sweater, and produce structures that seem, on larger scales, to approximate regions of spacetime. Most convincing of all, loop researchers have calculated the allowed areas of such surfaces of space. And just as you can have one electron or two electrons or 202 electrons, but you can't have 1.6 electrons or any other fraction, the calculations show that surfaces can have areas that are one square Planck-length, or two square Planck-lengths, or 202 square Planck-lengths, but no fractions are possible. Once again, this is a strong theoretical clue that space, like electrons, comes in discrete, indivisible chunks.[11]

If I were to hazard a guess on future developments, I'd imagine that the background-independent techniques developed by the loop quantum gravity community will be adapted to string theory, paving the way for a string formulation that is background independent. And that's the spark, I suspect, that will ignite a third superstring revolution in which, I'm optimistic, many of the remaining deep mysteries will be solved. Such developments would likely also bring spacetime's long story full circle. In earlier chapters, we followed the pendulum of opinion as it swung between relationist and absolutist positions on space, time, and spacetime. We asked: Is space a something, or isn't it? Is spacetime a something, or isn't it? And, over the course of a few centuries' thought, we encountered differing views. I believe that an experimentally confirmed, background-independent union between general relativity and quantum mechanics would yield a gratifying resolution to this issue. By virtue of the background independence, the theory's ingredients might stand in some relation to one another, but with the absence of a spacetime that is inserted into the theory from the outset, there'd be no